

Answer Key:

FINAL EXAM, VERSION 1
CSci 127: Introduction to Computer Science
Hunter College, City University of New York
Spring 2025

1. (a) What will the following Python code print:
- ```
mon_s = "January-February-March-April-May-June"
months = mon_s.split('-')
print(len(months), "months")
print("Last month is", months[-1])
short = [mo[:1] for mo in months]
mess = short[-1]
print("Short is:", short)
firsts = {}
for s in short:
 if s in firsts:
 firsts[s] = firsts[s]+1
 else:
 firsts[s] = 1
print("Months with A:", firsts['A'])
print("Months with J:", firsts['J'])
```

**Answer Key:**

```
6 months
Last month is June
Short is: ['J', 'F', 'M', 'A', 'M', 'J']
Months with A: 1
Months with J: 2
```

- (b) Consider the following shell commands:

```
$ ls
hello.cpp p1_hello.py p2_triangle.py
$ pwd
/tmp/final/v1
```

Assuming the commands below are run sequentially, what is the output after each has run:

- i. 

```
$ mv hello.cpp p1.cpp
$ ls
```

**Answer Key:**

```

p1_hello.py p1.cpp p2_triangle.py

$ mkdir pyprogs
ii. $ mv *.py pyprogs
$ ls

```

**Answer Key:**

```

p1.cpp pyprogs
$ cd pyprogs
iii. $ echo "Current directory:"
$ pwd

```

**Answer Key:**

```

Current directory:
/tmp/final/v1/pyprogs

$ cd ..
$ mkdir old_files
iv. $ cp p1.cpp old_files
$ echo "Count is:"
$ ls | wc -l

```

**Answer Key:**

```

Count is:
2

```

2. (a) For each question, **check all that apply**:

**Answer Key:**

- i. What color is `tom` after this command? `tom.color("#AA0000")`?
 

|                                |                                |                               |                                         |                               |
|--------------------------------|--------------------------------|-------------------------------|-----------------------------------------|-------------------------------|
| <input type="checkbox"/> white | <input type="checkbox"/> green | <input type="checkbox"/> gray | <input checked="" type="checkbox"/> red | <input type="checkbox"/> blue |
|--------------------------------|--------------------------------|-------------------------------|-----------------------------------------|-------------------------------|
- ii. What is the binary number equivalent to the decimal number 18?
 

|                                |                                |                                           |                                |                                |
|--------------------------------|--------------------------------|-------------------------------------------|--------------------------------|--------------------------------|
| <input type="checkbox"/> 00111 | <input type="checkbox"/> 01001 | <input checked="" type="checkbox"/> 10010 | <input type="checkbox"/> 10111 | <input type="checkbox"/> 11110 |
|--------------------------------|--------------------------------|-------------------------------------------|--------------------------------|--------------------------------|
- iii. Which of the **binary numbers** below are smaller than the decimal number 9?
 

|                                        |                                         |                               |                               |                               |
|----------------------------------------|-----------------------------------------|-------------------------------|-------------------------------|-------------------------------|
| <input checked="" type="checkbox"/> 10 | <input checked="" type="checkbox"/> 101 | <input type="checkbox"/> 1010 | <input type="checkbox"/> 1111 | <input type="checkbox"/> none |
|----------------------------------------|-----------------------------------------|-------------------------------|-------------------------------|-------------------------------|
- iv. Select the **smallest** hexadecimal number:
 

|                             |                             |                             |                                        |                             |
|-----------------------------|-----------------------------|-----------------------------|----------------------------------------|-----------------------------|
| <input type="checkbox"/> AA | <input type="checkbox"/> 31 | <input type="checkbox"/> 2C | <input checked="" type="checkbox"/> 1F | <input type="checkbox"/> FF |
|-----------------------------|-----------------------------|-----------------------------|----------------------------------------|-----------------------------|
- v. Which of the **hexadecimal numbers** below are larger than the decimal number 20?
 

|                            |                            |                                        |                                        |                               |
|----------------------------|----------------------------|----------------------------------------|----------------------------------------|-------------------------------|
| <input type="checkbox"/> A | <input type="checkbox"/> F | <input checked="" type="checkbox"/> 19 | <input checked="" type="checkbox"/> 5A | <input type="checkbox"/> none |
|----------------------------|----------------------------|----------------------------------------|----------------------------------------|-------------------------------|

- (b) After executing the Python code, write the name of the turtle:

i. which is red:

```
import turtle
ellie = turtle.Turtle()
turtle.colormode(1.0)
ellie.color(0.0, 0.0, 1.0)
fatima = turtle.Turtle()
turtle.colormode(255)
fatima.color(255, 0, 0)
guo = turtle.Turtle()
guo.color("#EFEFEF")
hector = turtle.Turtle()
hector.color("#009999")
```

**Answer Key:** fatima

ii. which is blue-green:

**Answer Key:** hector

iii. which is blue:

**Answer Key:** ellie

iv. which is gray:

**Answer Key:** guo

(c) Consider the code:

**Answer Key:**

```
(i) 1 mess == ""
(ii) 2 while mess == ""
3 mess = input('Enter non-empty string: ')
4 print(mess)
```

The answer should include:

- Mark line 1 with a “(i)”.
- In line 1, circle the == (should be =).
- Mark line 2 with a “(ii)”.
- At the end of line 5, box the space/parenthesis at the end of the line (where the missing colon should be).

i. **Circle** the code above and mark line with (i) that caused this error:

```
line 1: mess == ""
          ~~~~
```

NameError: name 'mess' is not defined

Write the code that would fix the error:

**Answer Key:**

```
mess = ""
```

ii. **Box** the code above and mark line with (ii) that caused this error:

```
line 2: while mess == ""
          ^
```

SyntaxError: expected ':'

Write the code that would fix the error:

**Answer Key:**

```
while mess == "":
```

3. (a) What is the value (True/False) of out:

```
in1 = False
```

- i. `in2 = True`

```
out = in1 and in2
```

**Answer Key:**

```
out = False
```

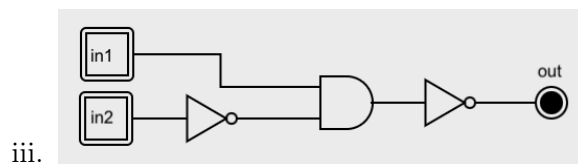
```
in1 = False
```

- ii. `in2 = False`

```
out = not in2 or (in2 and not in1)
```

**Answer Key:**

```
out = True
```



```
in1 = False
```

```
in2 = True
```

**Answer Key:**

```
out = True
```

- (b) Fill in the values to yield the output:

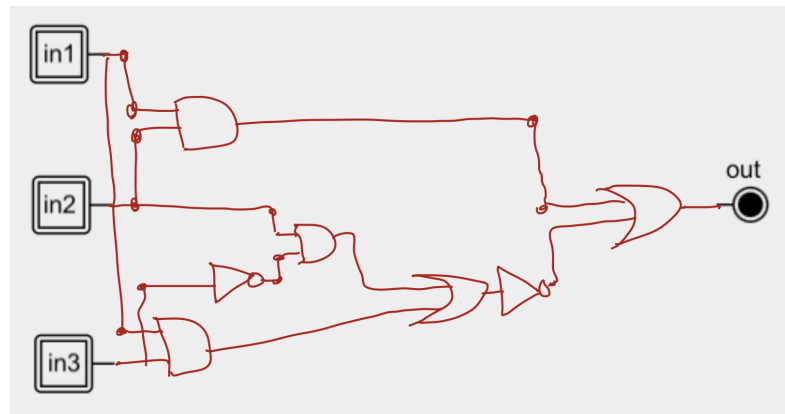
|          |                          |
|----------|--------------------------|
| in1 =    | <b>Answer Key:</b> True  |
| i. in2 = | <b>Answer Key:</b> False |

```
out = in1 and (not in1 or in2)
```

- (c) Design a circuit that implements the logical expression:

```
(in1 and in2) or not ((in1 and in3) or (in2 and not in3))
```

**Answer Key:**


$$v1: (in1 \text{ and } in2) \text{ or not } ((in1 \text{ and } in3) \text{ or } (in2 \text{ and not } in3))$$

4. (a) Draw the output for the function calls:

```
i. ramble(tim,0)
```

**Answer Key:**

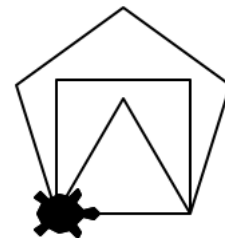


```
import turtle
tim = turtle.Turtle()
tim.shape("turtle")

def ramble(t,side):
    if side < 3:
        t.stamp()
    else:
        for i in range(side):
            t.forward(50)
            t.left(360/side)
        ramble(t,side-1)
```

- ii. `ramble(tim,5)`

**Answer Key:**



- (b) For the following code:

```
def v4(antonio, lola):
    if antonio + lola < 10:
        return antonio
    else:
        return -1
```

```
def start():
    jack = 5
    dandan = 20
    kate = v4(jack,dandan)
    return kate
```

- i. What are the formal parameters for `v4()`:

**Answer Key:** antonio, lola

- ii. What are the formal parameters for `start()`:

**Answer Key:** None

- iii. What value does `start()` return:

**Answer Key:** -1

5. Write a function `unique_visitors()` that takes a list of 8-digit strings and returns the number of unique strings that occur. For example:

```
ids = ['12345678', '11223344', '12312323', '12345678']
unique_visitors(ids)
```

would return 3 since there are 4 entries but the first and fourth entries are duplicates of each other.

|                    |                   |                                 |
|--------------------|-------------------|---------------------------------|
| <b>Answer Key:</b> | <b>Libraries:</b> | No additional– just core Python |
|                    | <b>Input:</b>     | list of 8-digit strings         |
|                    | <b>Output:</b>    | number of unique IDs            |

**Design Pattern:**

**Answer Key:**

☐ Accumulator   ☐ Max/Min   ☒ Finding Duplicates   ☐ Searching

**Principal Mechanisms** (select all that apply):

**Answer Key:**

☒ Single Loop   ☐ Nested Loop   ☒ Conditional (if/else)   ☐ Recursion  
☐ Indexing/slicing   ☒ Dictionary   ☐ List Comprehension   ☐ Regular Expressions

**Process** (as a concise and precise LIST OF STEPS / pseudocode):

(Assume libraries have already been imported.)

**Answer Key:**

- Set up an empty dictionary, `new_dict`.
- For each ID in the ID list:
- Check if the ID is in the dictionary.
- If it is, increment the count
- If it isn't, add ID with value 1 to the dictionary.
- Return the length of the dictionary, `new_dict`.

6. Fill in the Python program that will:

- prompt the user for the name of a CSV file,
- prompt the user for the name of a column in that CSV file,
- print out the maximum value of the column,
- print out the average value of the column, and
- displays a plot of the column entered (with "Year" as the x-axis).

**Answer Key:**

```
#Import the libraries for data frames and displaying images:
import pandas as pd
import matplotlib.pyplot as plt

#Prompt user for file name:
fileName = input('Enter file name: ')

#Prompt user for column name:
col = input('Enter column name: ')

df = pd.read_csv(fileName)

#Compute maximum value of the column:
M = df[col].max()
print("Maximum of column", col, "is", M)

#Compute average value of the column:
ave = df[col].mean()
print("Average of column", col, "is", ave)

#Display a plot of "Year" vs. column entered by user:
df.plot(x = "Year", y = col)
plt.show()
```

7. Write a complete Python program that

- asks the user for the name of a .png (image) file and
- prints the number of pixels that are very purple (the fraction of red and the fraction of blue are both above 0.75 and the fraction of green is below 0.25).

**Answer Key:**

```
#Count number of purple pixels in an image

#Import the packages for images and arrays:
import matplotlib.pyplot as plt
```

```

import numpy as np

fileName = input('Enter file name: ')
img = plt.imread(fileName)    #Read in image
count_purple = 0              #Number of pixels that are purple

#For every pixel:
for i in range(img.shape[0]):
    for j in range(img.shape[1]):
        if (img[i,j,0] > 0.75) and (img[i,j,1] < 0.25) and (img[i,j,2] > 0.75):
            count_purple = count_purple + 1

print("Purple count is", count_purple)

```

8. (a) Consider the following MIPS program:

```

ADDI $s0, $zero, 1
ADD $s1, $s0, $s0
ADD $s2, $s1, $s0
SUB $s3, $s1, $s2

```

After the program runs, what is the value stored in:

| \$s1 register | \$s2 register | \$s3 register  |
|---------------|---------------|----------------|
| Answer Key: 2 | Answer Key: 3 | Answer Key: -1 |

- (b) Consider the MIPS code:

```

1  ADDI $sp, $sp, -6
2  ADDI $t0, $zero, 65
3  ADDI $s2, $zero, 75
4  SETUP: SB $t0, 0($sp)
5  ADDI $sp, $sp, 1
6  ADDI $t0, $t0, 2
7  BEQ $t0, $s2, DONE
8  J SETUP
9  DONE: ADDI $t0, $zero, 0
10 SB $t0, 0($sp)
11 ADDI $sp, $sp, -5
12 ADDI $v0, $zero, 4
13 ADDI $a0, $sp, 0
14 syscall

```

Answer Key:



|                                                                            |                                                                                            |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| i) How many characters are printed?                                        | <b>5</b>                                                                                   |
| ii) What is the first character printed?                                   | <b>A</b>                                                                                   |
| iii) What is the whole message printed?                                    | <b>ACEGI</b>                                                                               |
| iv) Detail the changes needed to the code to print the message in reverse: | <b>Line 2: Start t0 at 73.<br/>Line 3: Start s2 at 63.<br/>Line 6: Subtract 2 from t0.</b> |

9. (a) What is the output

```
//Neil deGrasse Tyson
#include <iostream>
using namespace std;
int main()
{
    cout << "There is no "
          << "greater educ";
    cout << "ation\nthan one ";
    cout << "that is self-driven."
          << endl;
}
```

### Answer Key:

There is no greater education  
than one that is self-driven.

- (b) What is the output:

```
#include <iostream>
using namespace std;
int main()
{
    int year=1, bal=1000, expenses=200;
    while( bal > 0 ) {
        cout << "Year " << year
              << ": Balance: $"
              << bal << endl;
        bal = bal - expenses;
        year++;
    }
    return 0;
}
```

### Answer Key:

Year 1: Balance: \$1000  
Year 2: Balance: \$800  
Year 3: Balance: \$600  
Year 4: Balance: \$400  
Year 5: Balance: \$200

(c) What is the output:

```
#include <iostream>
using namespace std;
int main(){
    for (int i=0; i<5; i++){
        for(int j=0; j<5; j++){
            if ((i+j) % 2 == 0)
                cout<<"+";
            else
                cout<<"-";
        }
        cout << endl;
    }

    return 0;
}
```

### Answer Key:

```
+ - + - +
- + - + -
+ - + - +
- + - + -
+ - + - +
```

10. (a) Translate the C++ program into a **complete** Python program:

**C++ program:**

```
#include <iostream>
using namespace std;
int main()
{
    int num = 1;
    while ((num < 0) || (num%2 == 1))
    {
        cout << "Enter small even #:";
        cin >> num;
    }
    cout << "Your number: " << num;
    return 0;
}
```

**Python program:**

### Answer Key:

```
num = 1
while (num > 100) or (num % 2 == 1):
    num = int(input("Enter small even #: "))
print("Your number:", num)
```

(b) Write a C++ program that will ask for the time in 24 hour format (e.g. 2034 is 8:34pm) and, prints out “Morning Twilight” if the time is between 5am (e.g. 500) and 5:45am (e.g. 545), “Daylight” if the time is between 5:45am (e.g. 545) and 8pm (e.g. 2000) “Evening Twilight” if the time is between 8pm (e.g. 2000) and 8:30pm (e.g. 2030), and otherwise print “Night”

A sample run:

```
Enter time: 2015
Evening Twilight
```

**Answer Key:**

```
#include <iostream>
using namespace std;
int main()
{
    int time;
    cout << "Enter time: ";
    cin >> time;
    if ((500 < time) && (time < 545)) {
        cout << "Morning Twilight \n";
    }
    else if ((545 < time) && (time < 2000)){
        cout << "Daylight \n";
    }
    else if ((2000 < time) && (time < 2030)){
        cout << "Evening Twilight \n";
    }
    else {
        cout << "Night \n";
    }
    return 0;
}
```